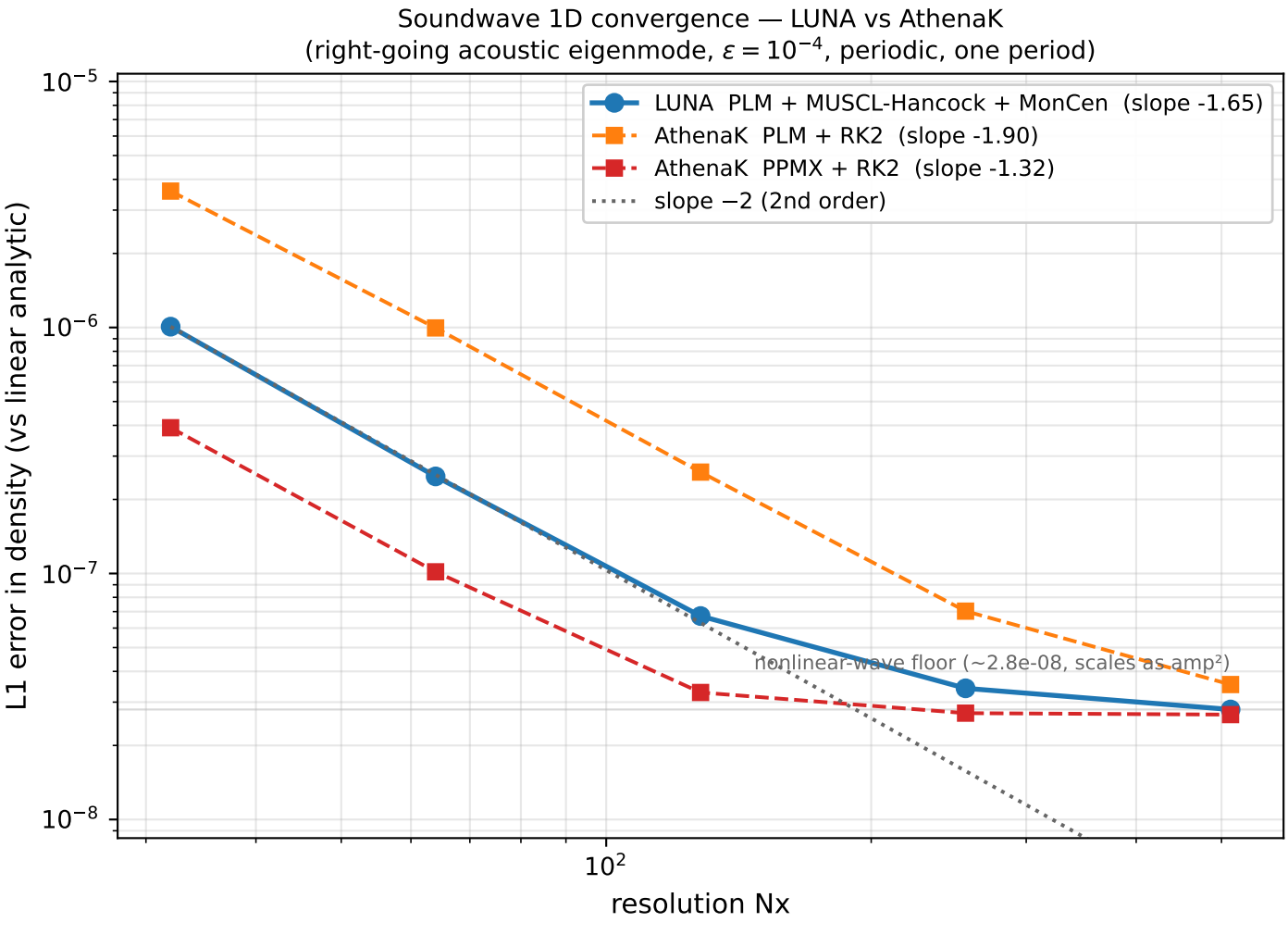
Soundwave 1D convergence — LUNA vs AthenaK
(right-going acoustic eigenmode, $\varepsilon = 10^{-4}$, periodic, one period)
LUNA PLM + MUSCL-Hancock + MonCen (slope -1.65)
AthenaK PLM + RK2 (slope -1.90)
AthenaK PPMX + RK2 (slope -1.32)
slope −2 (2nd order)
nonlinear-wave floor (~2.8e-08, scales as amp²)
L1 error in density (vs linear analytic)
resolution Nx
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{2}